

Wireshark Web Traffic Analysis

Network Traffic Capture, Analysis and Visualization

Wireshark | Microsoft Excel | Power BI | IntelliJ IDEA | GitHub

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1. Introduction and project purpose

This project is about analyzing the Web traffic of a local Web server with Wireshark. The objective of the project was to find out how communication happens in a network when one accesses a web page and what types of protocols and packets are being exchanged in the communication.

The project has been designed as a practical, learning-by-doing network analysis project. A simple website was developed and hosted locally using IntelliJ IDEA. The network traffic generated when they accessed the website was then recorded using Wireshark. They then used Wireshark to capture the network traffic that was generated while accessing the website. The captured packets were exported and analyzed in MS Excel, and a visual dashboard was created in Power BI.

2. Purpose of the project

This project was undertaken to gain practical experience in:

- Using Wireshark to capture network packets.
- Knowing how to communicate with HTTP and TCP.
- Detecting packets sent between a web browser and a local web server.
- Using Excel to analyze packet information.
- Building a basic Traffic dashboard with Power BI.
- Presenting network analysis results in a clear and understandable way.

3. Process of the project

1. Website Creation

A basic HTML website was developed using IntelliJ IDEA and then served locally with its integrated web server.

2. Traffic Capture

The traffic from the localhost website was captured using Wireshark and the Adapter for loopback traffic capture.

3. Traffic Analysis

The TCP, HTTP, and WebSocket traffic was analyzed using Wireshark filters. It was discovered that the following elements of the application layer existed.

4. Excel Analysis

The captured packets were exported from Wireshark in CSV format and then analyzed in a spreadsheet (Microsoft Excel). Statistics of packet lengths and protocol numbers were computed.

5. Power BI Dashboard

Data was analyzed and presented in Power BI using a basic network traffic dashboard.

6. GitHub Portfolio

The website, packet capture, CSV file, the Excel workbook, the Power BI dashboard, and screenshots were uploaded to GitHub to build a complete project portfolio.

4. Key Results

The analysis identified:

- TCP: 323
- HTTP: 48
- WebSocket: 14
- Maximum packet length: 2457 bytes
- Limit the maximum packet length: 2,590 bytes
- Rate: 82.34 Mbps

In the analysis, the authors illustrated how to use Wireshark to view actual network communications and how to use Excel and Power BI to organize and present the data.

5. Conclusion

This project gave hands-on experience in capturing and visualizing network traffic. It showed the complete procedure from making a local website to capturing the website traffic with Wireshark, analyzing the data with Excel, and visualizing it with Power BI.